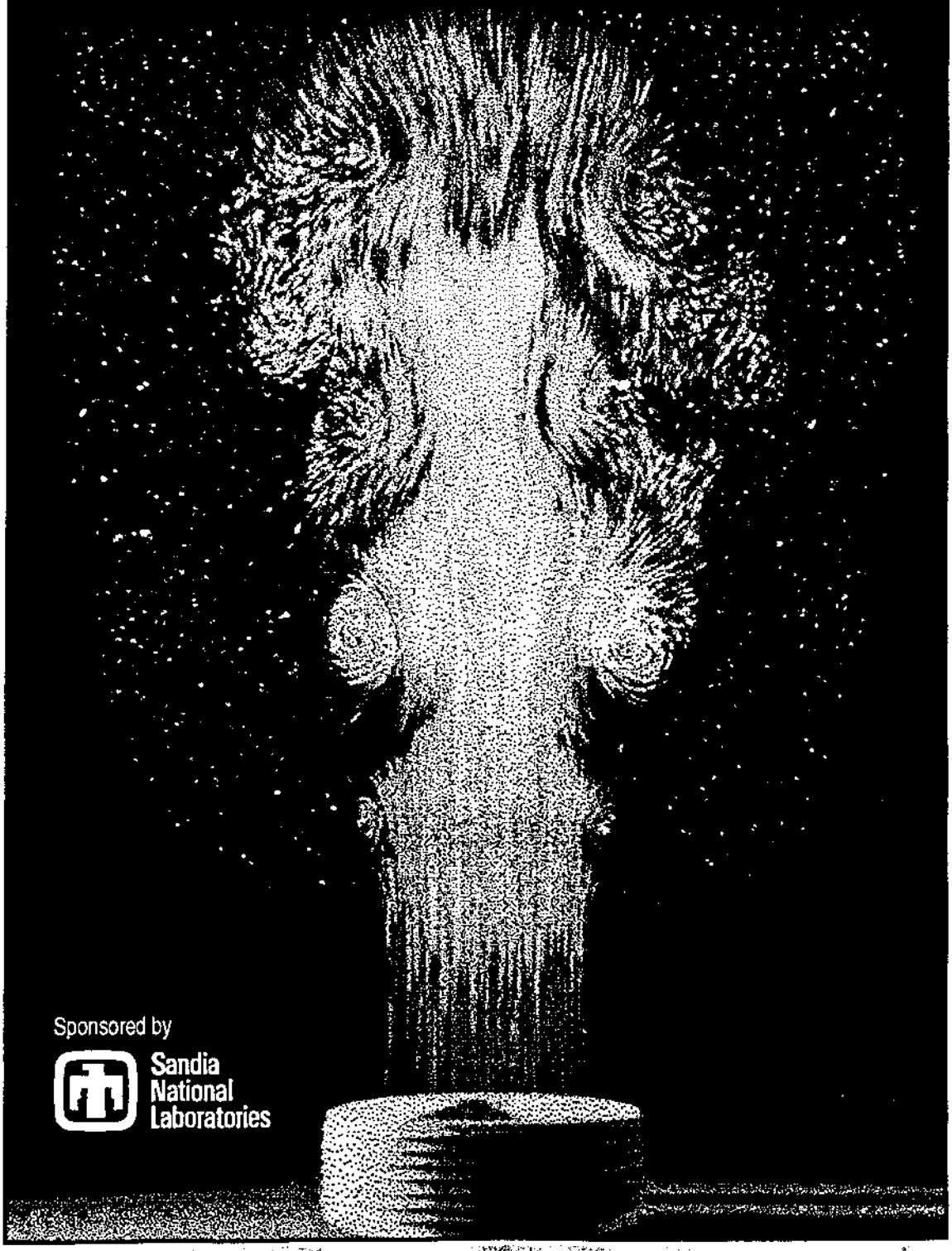
Forum on Vortex Methods for Engineering Applications

Albuquerque, New Mexico • February 22-24, 1995



Sponsored by

Sandia National Laboratories

Restaurant Location Map ii

Basic Issues:

Velocity Boundary Conditions for Vorticity Formulations of the Incompressible Navier-Stokes Equations, *Steve N. Kempka* 1

Three-Dimensional Vortex Particle and Panel Methods: Fast Tree-Code Solvers with Active Error Control for Arbitrary Distributions/Geometries, *Gregoire S. Winckelmans* 23

Marco Vortex Dynamics: A Vortex Dynamics Model with Inviscid Vortex Merging, *David M. Zielke* 45

A Lagrangian Numerical Method Fluid-Membrane Interaction in 2D and 3D, *G. Russo* 67

Viscous Diffusion

Viscous Diffusion of Vorticity in Unsteady Wall Layers Using the Diffusion Velocity Concept, *James H. Strickland* 69

A Deterministic Vortex Sheet Method for Three-Dimensional Boundary Layers, *Peter S. Bernard* 85

Simulating Diffusion in Vortex Methods Using a Vorticity Redistribution Technique, *Shankar Subramaniam* 105

Vortex Computations of Wall Jet Flows, *Louis F. Rossi* 125

Vortex Dynamics Simulation of Premixed Turbulent Flame Motion, *Inge R. Gran* 149

Element Modeling

A Lagrangian Collocation Method for Vorticity Transport in Viscous Fluid Flows, *Jeffrey S. Marshall* 173

Solution of the Vorticity Equation on a Lagrangian Mesh Using Interpolation Triangulation, *John R. Grant & Stephen A. Huyer* 193

Incorporation of Boundaries for 2D Triangular Vorticity Element Methods, *Stephen A. Huyer & John R. Grant.* 211

Panel Session & Technical Discussions

Notes 227

Plenary Lecture

Turbulent Flame Motion in a Pancake Chamber via a Lagrangian, Two-Dimensional Vortex Dynamics Simulation, *Bill Ashurst* 229

Simulation

Numerical and Experimental Investigations of the Unsteady Axisymmetric Flow Over a Disk, *Hiroshi Higuchi* 277

Numerical Simulation of Coupling Effects in Multiphase Free Shear Layers, *Clayton T. Crowe* 299

Unsteady Flow Through 2D Channels with a Constriction, *G. C. J. Hofmans* 321

Vorticity and Industrial Interactions, *Pam Barr* 335

List of Attendees 347

Agenda 355

VORTEX COMPUTATIONS OF WALL JET FLOWS

LOUIS F. ROSSI

December 22, 1994

ABSTRACT. A new vortex method permits practical and effective simulations of moderate Reynolds number wall jet flows at a low computational cost. Furthermore, this scheme is sufficiently general to be applied to other unsteady flow problems involving complicated boundaries. While traditional vortex methods appear naturally suited for simulations of vorticity dominated flows, the transport of vorticity through viscous diffusion and the creation of vorticity by solid boundaries require additional consideration since such mechanisms are not easily expressed in the usual Lagrangian framework. This new vortex method adaptively captures diffusive effects and satisfies the appropriate boundary conditions. Also, a new scheme merges overlapping computational elements, decreasing the problem size while inducing a controllably small error. The resulting vortex method achieved quantitative and qualitative agreement with laboratory wall jet experiments.

1. INTRODUCTION

The wall jet is a paradigm for vortex method solutions to flows around complex boundaries because both analytical and experimental studies are available for comparison with numerical simulations. The wall jet, a two-dimensional flow driven by a jet blown tangentially along a wall, is a challenging numerical problem for any method for a variety of reasons stemming from the wall jet's geometry. Since the flow is driven by narrow, isolated concentrations of vorticity in an unbounded domain, the vortex method would appear to be an ideal technique for capturing flow characteristics. In unbounded regions, vortex methods are more suited for the numerical investigation of flows with freely advecting regions of vorticity than fixed grid methods such as finite difference or spectral schemes because they dedicate computation effort only in regions containing vorticity. However, vortex methods suffer many shortcomings unique to grid free methods and vorticity formulations. Some examples of challenges for investigators using vortex methods include implementing vorticity boundary conditions and capturing diffusive effects. Since wall jet flows are driven by vorticity which diffuses from the wall and interacts with the jet in free space, the solutions to these problems yield accurate and effective simulations.

A planar wall jet is a two-dimensional flow driven by a jet blowing tangentially along a wall. In wind tunnel and water tank experiments, the flow is observed to remain two-dimensional far downstream hence a two-dimensional simulation would accurately capture the qualities of a full three-dimensional realization. In 1956, Glauert published a self-similar boundary layer solution for the wall jet, but this flow is known to be linearly unstable [7, 5]. This instability, observed

in laboratory experiments and numerical simulations, is characterized by the formation of dipoles that slowly evolve while moving with the flow. Far downstream, the dipoles depart from the solid wall causing the boundary layer to separate ejecting a familiar mushroom shaped dipole.

These vortex simulations of wall jets augment and enhance existing knowledge of wall jets obtained analytically and experimentally. First, it is difficult to observe and measure the initial wall jet formation in a wind tunnel or water tank. A numerical simulation is a natural choice for measuring transient effects. Next, the evolution of the coherent dipole structures are difficult if not impossible to study with physical experiments while numerical simulations provide access to the entire flow field. Finally, the numerical and physical experiments complement each other by verifying velocity and streakline measurements for common parameter values.

To simulate wall jets, a general scheme for constraining flow around a combination of flat plates is used. The nozzle and wall is decomposed into a series of these plates with specified flux and circulation boundary conditions. Solid walls impose zero flux and circulation, but it is also possible to specify blowing and suction. In order to constrain the flow to satisfy these boundary conditions, one injects vorticity into the flow and adds a potential flow corresponding to integrably weak sources and sinks along the plates. The distribution of the injected vorticity is a first order correction based on the unsteady Rayleigh problem [2]. While velocity boundary conditions are expressed at each point on the boundary independent of the flow, vorticity boundary conditions are fundamentally different in that the rate of vorticity injection into a bounded region is a function of the entire flow [1]. The new scheme is consistent with this result in that global constraints determine the potential solution and vorticity to be injected. Thus, the slip and normal flow conditions can be satisfied simultaneously. This approach is fundamentally different from that of Chorin, Sethian and others where the dual boundary conditions are never satisfied at the same time and vorticity is generated locally rather than globally [4, 16]. Thus, the new method captures many important properties of the bounded Navier-Stokes equations and is general enough for a wide range of complicated geometries.

This new vortex method captures the diffusive term using a corrected core spreading method. The vortex field is expressed as Gaussian vortex elements which convect freely with the flow. However, unlike other methods, the core sizes grow slowly in time to capture the diffusive term of the Navier-Stokes equations. Since the trajectories and core sizes are integrated in time simultaneously, there is no operator splitting as in the vortex methods of Chorin and others [3]. However, there is a latent inconsistency in core spreading that must be corrected [8, 13]. For this method, a fission process replaces an element which has spread beyond a prescribed length scale with many thinner elements. The fission process induces a controllably small error while allowing vorticity to diffuse in a frame moving with the flow. The new method is fully deterministic and hence does not require ensemble averaging as do more traditional random walk methods. More importantly, the resulting method, core spreading corrected by vortex fission, converges to an exact solution to the Navier-Stokes equations while still maintaining a high computational efficiency [12].

For downstream calculations where the nozzle shape is no longer important, it is more practical to simulate unsteady perturbations to the known similarity solution. Rather than compute the entire solution with a vortex method, one can use Glauert's analytical solution and compute only the deviations as they grow in time. This problem analytically reduces to solving forced vorticity equations using a vortex method. This forcing term describes the rate of vorticity generation in the domain and can be approximating by adding vortex elements to the flow at regular time intervals.

To control the growth of the problem size, it is possible to merge many overlapping elements into a single element, inducing a controllably small error. As vorticity diffuses into regions of irrotational flow and sheds off solid boundaries, the number of computational elements grows without a physical mechanism to mediate the process, except perhaps removing elements as they drift away from some domain of interest. Merging elements together is a new way to decrease an unnecessarily large problem size. Using this merging scheme, it is possible to control the problem size, yielding greater computational efficiency without sacrificing accuracy.

The resulting vortex simulation using the new methods of satisfying boundary conditions and capturing viscous effects succeeded in simulating wall jet flows matching a variety of experimental conditions. Regularly injected passive elements generated streaklines for comparison with water tank flow visualizations. The numerical simulations produced some of the first measurements of the initial wall jet transient and illuminated its later role in the formation of the "steady" wall jet. In all, the new vortex method produced information about wall jets with moderate computational resources.

2. A Corrected Core Spreading Method

Core spreading was one of the earlier techniques used to capture the viscous term in the vorticity equation,

$$\frac{D\omega}{Dt} = \nu \nabla^2 \omega \tag{1}$$

for the two-dimensional, incompressible Navier-Stokes equations where $\frac{D}{Dt} = \partial_t + (\vec{u} \cdot \nabla)$. All vortex methods express the vorticity field as a linear combination of basis functions,

$$\omega(\vec{x}) = \sum_{i=1}^{N} \gamma_i f(\vec{x} - \vec{x}_i, t), \tag{2}$$

where $\vec{x}_i$ is the position and γ_i is the circulation of each basis function. The basis functions rigidly advect with the velocity computed at the centroid via the Biot-Savart law applied to each vortex element. To capture viscous effects, core spreading methods adjust the basis function shape, f. For example, using Gaussian basis functions with width σ_i for each element,

$$\omega(\vec{x}) = \sum_{i=1}^{N} \frac{\gamma_i}{4\pi\sigma_i^2} \exp\left(-\frac{|\vec{x} - \vec{x}_i|^2}{4\sigma_i^2}\right), \tag{3}$$

one can take advantage of the fact that gaussians are self-similar solutions to the diffusion equation. The resulting core spreading method is described by the following system of ODEs:

$$\dot{\vec{x}}_i = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \sum_{j=1}^{N} \frac{\gamma_i}{2\pi} \frac{\vec{x}_i - \vec{x}_j}{|\vec{x}_i - \vec{x}_j|^2} \left[1 - \exp\left(-\frac{|\vec{x}_i - \vec{x}_j|^2}{4\sigma_i^2}\right)\right], \tag{4}$$

$$\dot{\sigma}_i^2 = \nu. \tag{5}$$

In addition to being fully deterministic, this system captures equation (1) without splitting the evolution equation into convective and diffusive parts *if* the core sizes are controllably small.

While a core spreading method by itself is *not* consistent with the equation (1) , a vortex splitting scheme, called fission, can correct the method. A cursory explanation for this inconsistency is that

the core width, σ_i, of every element at time t is at least $\sqrt{\nu t}$ and cannot be controlled by numerical parameters. This lower bound on core size implies that convective errors induced by the vortex method cannot be made arbitrarily small. A more detailed explanation of the inconsistency in simple core spreading schemes is found in a letter by C. Greengard [8].

The corrected core spreading vortex method (CCSVM) enforces an arbitrary upper bound on core width through a process called vortex fission. This process replaces a single vortex element of width l with several elements of width αl so that the new configuration of elements closely approximates the original element. The parameter $\alpha \in (0,1]$ controls the accuracy of the fission process. The most intuitive fission procedure uses a configuration of four elements. By comparing zeroth, first and second moments of vorticity before and after fission, the precise placement of elements is completely determined: The original vortex,

$$\frac{\gamma}{4\pi\sigma^2}\exp\left(-\frac{\|\vec{x}\|^2}{4\sigma^2}\right), \tag{6}$$

is replaced by four elements,

$$\sum_{i=1}^{4}\left(\frac{\gamma}{4}\right)\frac{1}{4\pi(\alpha\sigma)^2}\exp\left[-\frac{\|\vec{x}+\vec{r}_i\|^2}{4(\alpha\sigma)^2}\right], \tag{7}$$

where $\vec{r}_i = \{\left[\begin{smallmatrix} r \\ 0 \end{smallmatrix}\right], \left[\begin{smallmatrix} -r \\ 0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0 \\ r \end{smallmatrix}\right], \left[\begin{smallmatrix} 0 \\ -r \end{smallmatrix}\right]\}$, $r = 2\sigma\sqrt{1-\alpha^2}$. As $\alpha \to 1$, vortex fission becomes more accurate but more frequent. The period of time over which fission occurs is called the viscous timestep,

$$\Delta T_\nu = \frac{l^2(1-\alpha^2)}{\nu}. \tag{8}$$

The fission process is related to random walk techniques in that n repeated fission events correspond to 4^n particles tracing all possible paths of length n from a fixed point on a discrete unit grid. The stochastic process described by 4^n particles undergoing a 2D random walk from a central point rapidly converges to the same distribution.

The fission process necessarily increases the problem size adaptively to maintain accuracy as vorticity diffuses across streamlines. There are many flows for which the increase in density of the vortex elements is not efficient. In these situations, it is possible to merge many nearby elements into a single element, effectively decreasing the problem size while maintaining accuracy. A more complete discussion of fusion techniques as well as the accuracy and stability of CCSVM is beyond the scope of this paper but can be found in [12, 13].

3. Constraining the Flow to Satisfy the Full Boundary Conditions

While vortex methods are efficient and accurate in many calculations, expression of analytical vorticity boundary conditions is a major shortcoming in computing wall bounded flows. The standard approach is to rely upon the velocity boundary conditions, no-slip and no normal flow, to compute the flow correctly. Since most vortex methods rely upon operator splitting, most researchers satisfy the no-slip and no normal flow boundary conditions separately [3, 9, 15, 1]. The shortcomings of these schemes motivates the alternative method discussed in this section.

In his original paper on slightly viscous flows, Chorin observed that any slip along a boundary can be corrected by the injection of an appropriate amount of vorticity [3]. Of course, it is difficult to determine where to inject vorticity save that it should correspond to a thin layer near the boundary.

The conventional strategy is to add an irrotational flow to satisfy no normal flow conditions on the boundary. This irrotational flow is determined by the normal flow component on the boundary during the inviscid step [16]. Then, the local slip along the boundary is computed to determine how much vorticity to inject into the flow. The injection procedure is usually coupled to the diffusion scheme so that vorticity diffuses out of the boundary such as random walks in the diffusive part of the split-step method.

A vortex method, such as CCSVM, that does not rely upon operator splitting provides an opportunity to satisfy both boundary conditions simultaneously. One shortcoming of the more traditional splitting approach is that flow field never satisfies the full boundary conditions at the same time. (In fact, the author has not found in literature any reference to an investigator measuring slip and normal flow velocities at the boundary when using traditional methods, and it is not clear that the procedure will converge to the correct zero flow boundary condition in some limit.) Another indication that the conventional injection technique of Chorin and others will fail is that the analytic formulations of vorticity boundary conditions assert that vorticity flux from the boundary is nonlocal [1, 10, 11]. The technique proposed in this paper satisfies both the no-slip and no-normal flow boundary conditions simultaneously. These constraints necessarily require nonlocal adjustments to the injected vorticity as well as the potential flow.

The flow field is constrained to satisfy the full boundary conditions by adding a potential flow and a rotational flow simultaneously. To apply these constraints, the boundary is divided into finite boundary segments, and the superimposed flows are chosen so that the integrated velocity on the boundary segments is zero. The potential flow is created through the use of a new computational unit called a "line source." A rotational flow is added to the domain by adding vortex elements near the boundary segments to approximate a thin boundary layer. With suitable superposition of boundary layers and sources along the boundary, the flow field approximates the full Navier Stokes equations with zero flow on solid boundaries.

3.1. Line Sources and the No Normal Flow Condition. The superposed potential flow is induced by computational elements called line sources. Line sources are infinitesimal mass point sources spread along a continuous line segment. Locally, each line source is balancing normal flow across a boundary segment with a source or sink. Of course, this could also be accomplished by a point source, but line sources have the advantage that the point source singularity in the induced velocity field is considerably weakened.

Every boundary segment is associated in location and orientation with a line source. Each line source has a length, l_i, inward normal $\hat{n}_i = \left[\begin{smallmatrix} n_i^x \\ n_i^y \end{smallmatrix}\right]$ and position $\vec{x}_i = \left[\begin{smallmatrix} x_i \\ y_i \end{smallmatrix}\right]$. Consequently, the tangent vector shall be defined as $\hat{t}_i = \left[\begin{smallmatrix} n_i^y \\ -n_i^x \end{smallmatrix}\right]$. Using the center of each segment as an origin, $\hat{n}_i$ and $\hat{t}_i$ define a local coordinate system for the ith boundary segment. (Indices will not be used when interaction between segments is not important.) Consequently, velocities in the local reference frame shall be denoted $\vec{\bar{u}} = \left[\begin{smallmatrix} \bar{u} \\ \bar{v} \end{smallmatrix}\right]$.

A uniform distribution of sources placed along a boundary segment induce the local velocity,

$$\tilde{u} = \frac{m}{2\pi}\int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{\tilde{x}-s}{(\tilde{x}-x)^2+\tilde{y}^2}ds = \frac{m}{4\pi}\ln\left[\frac{(\tilde{x}+\frac{l}{2})^2+\tilde{y}^2}{(\tilde{x}-\frac{l}{2})^2+\tilde{y}^2}\right], \tag{9}$$

$$\tilde{v} = \frac{m}{2\pi}\int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{\tilde{y}}{(\tilde{x}-x)^2+\tilde{y}^2}ds = \frac{m}{2\pi}\left[\tan^{-1}\left(\frac{\tilde{x}+\frac{l}{2}}{\tilde{y}}\right) - \tan^{-1}\left(\frac{\tilde{x}-\frac{l}{2}}{\tilde{y}}\right)\right]. \tag{10}$$

The resulting potential flow field is finite in the vertical direction and singular though integrable in the direction tangent to the wall.

3.2. The Dreaded No-Slip Condition. The fact that vorticity is produced at boundaries and either sheds or diffuses into the flow has long been observed. Indeed, the no-slip condition and viscosity interact closely at the boundary to produce a vorticity flux out of the boundary. In two dimensions, viscosity both produces vorticity at the walls and diffuses vorticity away from the walls. Further physical insight is required to specify the amount and distribution of vorticity to inject near the boundaries.

Since the amount of vorticity injected at each wall segment is determined globally for reasons discussed earlier, it is the distribution of vorticity near the wall which must be determined a priori and applied to the vortex method. The general algorithm for satisfying the boundary conditions indicates a time scale over which the vorticity distribution must be modeled:

1. Vortices advect and diffuse freely over a short distance $\Delta\vec{x}_i = \vec{u}(\vec{x}_i)\Delta t$ where $\vec{u}$ is a velocity field including Biot-Savart and spreading interactions from equations (4,5) and possibly previous velocity fields to yield more accurate numerical trajectory integration. Presumably, $\vec{u}$ satisfies the specified boundary conditions.
2. After the vortex elements move and spread, the velocity components normal and tangential to each boundary segment is used to determine a potential flow from line sources and to inject vortices near the boundary so that $\vec{u}$ once again satisfies the specified boundary conditions.

When the boundary conditions are enforced over a suitably short time interval, Δt, the dominant vorticity transport mechanism is viscous diffusion. Only this viscous transport needs to be modelled while the vortex method itself can handle more complicated vorticity evolution after elements are injected.

A crude but satisfactory model is found in the solution to the unsteady Rayleigh problem. Since $\nu\Delta t$ is very small compared to the length l of a typical boundary segment, the injected vorticity can be approximated by a thin region unidirectional flow above the solid boundary. In this case, equation (1) reduces to

$$\partial_t\omega(y,t) = \nu\partial_{yy}\omega(y,t) \tag{11}$$

with the initial condition $\omega(y,0) = -\delta(y)$. The solution to this problem over a short time period is

$$\omega(y) = -\frac{2u_0}{\sqrt{4\pi\nu\Delta t}}\exp\left(-\frac{y^2}{4\nu\Delta t}\right). \tag{12}$$

along with the condition that $u = u_0$ as $y \to \infty$ which is automatically satisfied with localized vortex elements. In this case, u_0 is not the local slip velocity as employed by some investigators

but rather, a parameter determined globally so that the boundary conditions are simultaneously satisfied everywhere.

There are many techniques for injecting vortex elements into the flow to approximately satisfy the no-slip condition including circulation exchange, elliptical blobs and vortex sheets [4, 17, 6, 9, 16]. Regardless of the technique, the distribution of this injected vorticity should have some physical relevance to the problem, and equation (12) serves an approximation for a locally flat boundary for small values of $\nu\Delta t$.

3.3. Simultaneous Boundary Conditions. As discussed earlier, each vortex blob and line source will induce some flux and some circulation on every other boundary element in addition to balancing flux and circulation locally. If there are B wall elements, there are $2B$ equations constraining the flow field. To solve this system, there are $2B$ parameters, B injected vortex formations and B line sources. Many vortices can be injected near a single wall element to generate a particular vorticity distribution but they induce a single free parameter. More particularly, many vortex elements are injected in rows along a given wall element to approximate equation (12). The total circulation, γ_i, of this configuration of vortices is a single parameter to be constrained by the boundary conditions. Since each line source is superimposed on a finite wall element, I shall denote the source strength, m_i, of each line element with the same subscript as the line element. That is, each wall element has a normal, $\hat{n}_i$, a tangent, $\hat{t}_i$ and a line source with strength m_i. For the remainder of this thesis, flux shall be the flux of material into the domain $\int \vec{u} \cdot \hat{n}_i$ and circulation will be oriented in the tangent direction $\int \vec{u} \cdot \hat{t}_i$.

To constrain the velocity field, one must compute the coefficients of the linear system

$$\sum_{j=0}^{B} m_j a_{ij} + \gamma_{in_j} b_{ij} = -f_i \tag{13}$$

$$\sum_{j=0}^{B} m_j c_{ij} + \gamma_{in_j} d_{ij} = -\Gamma_i \tag{14}$$

where a_{ij} is the flux induced by line source j across wall element i, d_{ij} is the circulation induced by vortex blob j on wall element i, and so forth. The net flux, f_i, is generated by vortex blobs in the system (which are not to be confused with the newly injected vortices whose strengths have yet to be determined) across element i. Likewise, Γ_i is the net induced circulation from vortex blobs. By adding an offset value to f_i, the wall elements blow or suck fluid into or out of the domain. Likewise, a circulation can be specified along boundary sections to simulate a slip velocity. One possible reason for enforcing a slip velocity would be to simulate the flow around a rotating cylinder.

3.4. Evaluating Vortex and Line Source Induced Fluxes and Circulations. Computing the flux and circulation induced by each vortex blob across each wall requires numerical integration because the integral of $\vec{u} \cdot \hat{n}$ from equation (4) has no known analytic solution over a finite domain. This is the case with most vortex method basis functions. Thus, the coefficients b_{ij} and d_{ij} as well as the right hand sides of equations (14,13) must be determined numerically. Analytic solutions exist for flux and circulation quantities induced by a line source along line segments. Thus, it is possible to compute a_{ij} and c_{ij} exactly.

In equation (13), the diagonal terms a_{ii} are equal to $\frac{1}{2}$ simply because half of the line source output passes into and half passes out of the domain. For equation (14), $c_{ii} = 0$ because the field induced by a line source is irrotational everywhere. The strategy for evaluating these integrals is to parameterize a curve along wall element j and compute the induced flux and circulation induced by element i. The lengths of these walls shall be referred to as l_j and l_i, respectively. Therefore, a parametric equation for the position of wall element j is

$$\vec{x}(s) = \vec{x}_j + s\hat{t}_j.$$

The linear transformation which will carry vectors in the standard basis to vectors in the local frame of wall element i is

$$Z = \begin{bmatrix} n_i^y & -n_i^x \\ n_i^x & n_i^y \end{bmatrix}.$$

Shifting the parametric equation into the local coordinates of wall i, yields

$$\tilde{\vec{x}}(s) = Z[(\vec{x}_j - \vec{x}_i) + s\hat{t}_j]. \tag{15}$$

Consequently,

$$\begin{aligned} \tilde{x} &= A + Bs, \\ \tilde{y} &= C + Ds, \end{aligned} \tag{16}$$

where

$$\begin{aligned} A &= (x_j - x_i)n_y^i - (y_j - y_i)n_x^i, \\ B &= n_y^i n_y^j + n_x^i n_x^j, \\ C &= (x_j - x_i)n_x^i + (y_j - y_i)n_y^i, \\ D &= n_x^i n_y^j - n_y^i n_x^j. \end{aligned} \tag{17}$$

In this local frame, allowing $\tilde{\vec{u}}$ to be the velocity field induced by line source i described by equations (9) and (10),

$$a_{ij} = \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tilde{\vec{u}} \cdot Z\hat{n}_j ds = \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} (-D\tilde{u} + B\tilde{v}) ds, \tag{18}$$

$$c_{ij} = \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tilde{\vec{u}} \cdot Z\hat{t}_j ds = \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} (B\tilde{u} + D\tilde{v}) ds. \tag{19}$$

The problem reduces to integrating logarithms, $\int \tilde{u}$, and arctangents, $\int \tilde{v}$, along quadratic paths.

Applying equation (9), the former becomes

$$\begin{aligned}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tilde{u} ds =& \frac{1}{4\pi}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln[(\tilde{x}+\frac{l_i}{2})^2+\tilde{y}^2]-\ln[(\tilde{x}-\frac{l_i}{2})^2+\tilde{y}^2]ds \\ =& \frac{1}{4\pi}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln[(A+Bs+\frac{l_i}{2})^2+(C+Ds)^2]- \\ & \ln[(A+Bs-\frac{l_i}{2})^2+(C+Ds)^2]ds \\ =& \frac{1}{4\pi}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln[(B^2+D^2)s^2+2(CD+B(A+\frac{l_i}{2})s+(A+\frac{l_i}{2})^2+C^2]- \\ & \ln[(B^2+D^2)s^2+2(CD+B(A-\frac{l_i}{2})s+(A-\frac{l_i}{2})^2+C^2]ds. \end{aligned} \tag{20}$$

Therefore, finding $\int \tilde{u}$ reduces to evaluating

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln[(B^2+D^2)s^2+2(CD+B\tilde{A})s+\tilde{A}^2+C^2]ds \tag{21}$$

where $\tilde{A}$ is either $A+\frac{l_i}{2}$ or $A-\frac{l_i}{2}$. In the general case, one would locate all roots of the quadratic polynomials and integrate with care. However, these quadratic polynomials merely represent the Euclidean distance from a point on wall element j to either the left or right endpoints of wall i, respectively. A root corresponds to a meeting of the two wall elements. In any realistic configuration, a point of contact will only occur when the left edge of one wall element touches the right edge of another wall element. Therefore, if there is a singularity, one can only expect to find it at the endpoints of the interval of integration. It is certain that such a root is a double root, as no line can pass through the same point twice.

Assuming that neither endpoint is a singularity, the integration of equation (20) is routine but tedious. First, it is simpler to specify temporary variables to ease the notation:

$$\begin{aligned} a =& B^2+D^2, \\ b =& 2(CD+B\tilde{A}), \\ c =& \tilde{A}^2+C^2. \end{aligned} \tag{22}$$

Then, equation (21) can be rewritten as

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln(as^2+bs+c)ds.$$

Integrating logarithms of quadratic polynomials requires knowledge of the sign of the discriminant.

In this case, $q = 4ac - b^2$ will evaluate to $4(\bar{A}D - BC)^2$ which is always positive.

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln(as^2 + bs + c)ds = (s + \frac{b}{2a}) \ln(as^2 + bs + c) - \frac{\sqrt{4ac - b^2}}{a} \tan^{-1}\left(\frac{2as + b}{\sqrt{4ac - b^2}}\right)\Bigg|_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tag{23}$$

While the cost of this computation may appear high, it is only computed once if the boundary is stationary, and the number of boundary elements is usually much smaller than the number of vortex elements.

Surprisingly, a singularity on the interval is simpler to compute than the preceding calculation. Without loss of generality, let us suppose that the right edge of element i ($s = \frac{l_i}{2}$) touches the left edge of element j ($s = -\frac{l_j}{2}$). In this case, the latter term of equation (20) would be evaluated using equation (23). But, the first term can be simplified.

In this case, the parameter value $s = -\frac{l_j}{2}$ is a double root of $as^2 + b_1 s + c_2$. So, equation (3.4) can be rewritten as

$$\begin{aligned}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln(as^2 + bs + c)ds &= \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln\left[a\left(s + \frac{l_j}{2}\right)^2\right] ds \\ &= 2\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \ln\left[\sqrt{a}\left(s + \frac{l_j}{2}\right)\right] ds \\ &= 2l_j[\ln(\sqrt{a}l_j) - 1].\end{aligned} \tag{24}$$

The same result is obtained if the root is located at $s = \frac{l_i}{2}$.

What remains is to integrate equation (10). In terms of the parameterized equations for $\bar{\bar{x}}$, this equation evolves into

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tilde{v} ds = \frac{m}{2\pi} \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \left[\tan^{-1}\left(\frac{A + Bs + \frac{l_i}{2}}{C + Ds}\right) - \tan^{-1}\left(\frac{A + Bs - \frac{l_i}{2}}{C + Ds}\right)\right] ds \tag{25}$$

for which, despite the fact that arctangent is a nice integrable function, there are many special cases which are enumerated below:

1. If $C = D = 0$ (or in other words $\tilde{y} = 0$), then $\int \tilde{v} = 0$ under the reasonable assumption that no wall elements overlap.
2. If $D = 0$ and $C \neq 0$, then equation (25) can be evaluated using

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tan^{-1}\left(\frac{\bar{A} + Bs}{C}\right) ds = \left(s + \frac{\bar{A}}{B}\right) \tan^{-1}\left(\frac{\bar{A} + Bs}{C}\right) - \frac{C}{2B} \ln\left[1 + \left(\frac{Bs + \bar{A}}{C}\right)^2\right]\Bigg|_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tag{26}$$

where $\bar{A}$ is either $A + \frac{l_i}{2}$ or $A - \frac{l_i}{2}$.

3. If the numerator and denominator have roots for the same value of s, it must necessarily occur at either endpoint of the region of integration because the numerator and denominator correspond to displacements from either the left or right endpoint in the local coordinates of wall element i. Without loss of generality, one can assume that the left endpoint of wall element j ($s = -\frac{l_j}{2}$) makes contact with the right edge of wall element i corresponding to the second term of equation (25). In this case, this second term is constant because wall element j radiates away from this edge. So, the endpoint singularity can be disregarded and the value of the integrand can be evaluated at any other value of s. Hence, the second term of equation (25), in this special case, is

$$\frac{m}{2\pi}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tan^{-1}\left(\frac{A+Bs-\frac{l_i}{2}}{C+Ds}\right) ds = \frac{ml_j}{2\pi}\tan^{-1}\left(\frac{A+(B-1)\frac{l_i}{2}}{C+D\frac{l_i}{2}}\right). \tag{27}$$

Here I choose to evaluate s at the right endpoint of wall j.

4. If the denominator has a root for a value of s which is not a root of the numerator, the integration can be carried out normally. Integration of equation (25) can be accomplished using integration by parts. Since the singularity occurs at $s = -\frac{C}{D}$, the integrating factor should be chosen to be $(s + \frac{C}{D})$.

$$\left(\int_{-\frac{C}{D}}^{\frac{l_j}{2}} + \int_{-\frac{l_j}{2}}^{-\frac{C}{D}}\right) \tan^{-1}\left(\frac{\tilde{A}+Bs}{C+Ds}\right) ds = \left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\tilde{A}+Bs}{C+Ds}\right)\Big|_{-\frac{C}{D}}^{\frac{l_j}{2}}\Big|_{-\frac{l_j}{2}}^{-\frac{C}{D}} -$$
$$\left(\int_{-\frac{C}{D}}^{\frac{l_j}{2}} + \int_{-\frac{l_j}{2}}^{-\frac{C}{D}}\right)\left(s+\frac{C}{D}\right)\left[1+\left(\frac{\tilde{A}+Bs}{C+Ds}\right)^2\right]^{-1}\left(\frac{BC-\tilde{A}D}{(C+Ds)^2}\right) ds. \tag{28}$$

Thus, the singularity is integrable.

$$\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \tan^{-1}\left(\frac{\tilde{A}+Bs}{C+Ds}\right) ds =$$
$$\left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\tilde{A}+Bs}{C+Ds}\right)\Big|_{-\frac{l_j}{2}}^{\frac{l_j}{2}} - \int_{-\frac{l_j}{2}}^{\frac{l_j}{2}}\left(s+\frac{C}{D}\right)\left[1+\left(\frac{\tilde{A}+Bs}{C+Ds}\right)^2\right]^{-1}\cdot\left[\frac{BC-\tilde{A}D}{(C+Ds)^2}\right] ds =$$
$$\left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\tilde{A}+Bs}{C+Ds}\right)\Big|_{-\frac{l_j}{2}}^{\frac{l_j}{2}} - \frac{BC-\tilde{A}D}{D}\int_{-\frac{l_j}{2}}^{\frac{l_j}{2}} \frac{(C+Ds)}{(C+Ds)^2+(\tilde{A}+Bs)^2} ds \tag{29}$$

Once again, the individual terms in the denominator correspond to displacements in orthogonal directions (see equation (16)). For this case, these two are not zero for the same value of s so this function has no singularities. Again, it is helpful to define temporary variables:

$$\begin{aligned} a &= B^2 + D^2, \\ b &= 2(\tilde{A}B + CD), \\ c &= \tilde{A}^2 + C^2. \end{aligned} \tag{30}$$

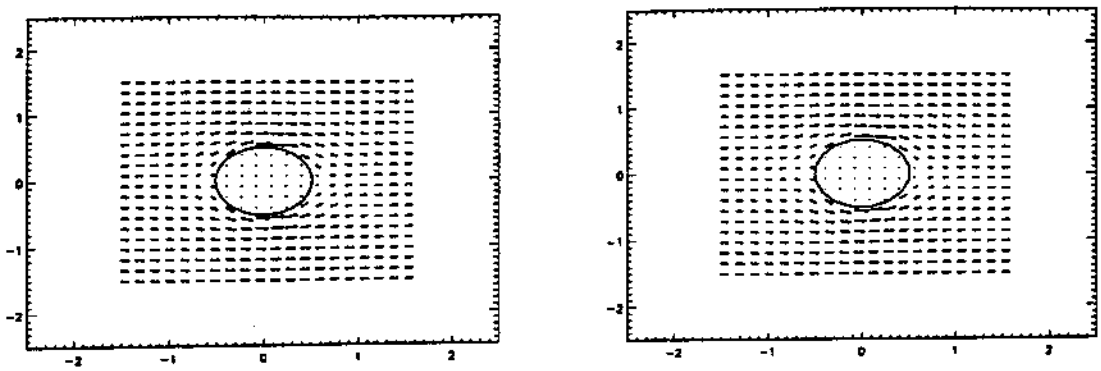


FIGURE 1. Potential flow around a cylinder (left) approximated by a 16-sided regular polygon (right).

Once again, the discriminant $q = 4ac - b^2$ has the good taste to be $q = 4(\bar{A}D - BC)^2$ and thus always positive.

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \tan^{-1}\left(\frac{\bar{A}+Bs}{C+Ds}\right) ds =$$
$$\left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\bar{A}+Bs}{C+Ds}\right)\Big|_{-\frac{1}{2}}^{\frac{1}{2}} - \frac{BC-\bar{A}D}{D}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{C+Ds}{as^2+bs+c}ds =$$
$$\left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\bar{A}+Bs}{C+Ds}\right)\Big|_{-\frac{1}{2}}^{\frac{1}{2}} - \left(\frac{BC-\bar{A}D}{D}\left(C-\frac{Db}{2a}\right)\frac{2}{\sqrt{q}}\tan^{-1}\left(\frac{2as+b}{\sqrt{q}}\right)\Big|_{-\frac{1}{2}}^{\frac{1}{2}} + \frac{BC-\bar{A}D}{2a}\ln(as^2+bs+c)\Big|_{-\frac{1}{2}}^{\frac{1}{2}}\right) \quad (31)$$

Without the temporary variables, the final result is

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \tan^{-1}\left(\frac{\bar{A}+Bs}{C+Ds}\right) ds = \left(s+\frac{C}{D}\right)\tan^{-1}\left(\frac{\bar{A}+Bs}{C+Ds}\right) + -$$
$$\frac{BC-\bar{A}D}{2(B^2+D^2)}\ln[(\bar{A}+Bs)^2+(C+Ds)^2]\Big|_{-\frac{1}{2}}^{\frac{1}{2}} -$$
$$\frac{BC-\bar{A}D}{|BC-\bar{A}D|}\left(\frac{C}{D}-\frac{\bar{A}B+CD}{B^2+D^2}\right)\tan^{-1}\left[\frac{(B^2+D^2)s+\bar{A}B+CD}{|\bar{A}D-BC|}\right]. \quad (32)$$

Now, all the vortex-boundary and boundary-boundary interactions are determined so that equation (13) and equation (14) can be solved. If the boundary does not move with time then vorticity is always injected in the same place near the boundary. Since only the right hand side of the linear system changes, this satisfaction of boundary conditions is computationally efficient because one would only have to decompose (QR, LU, etc.) the matrix once.

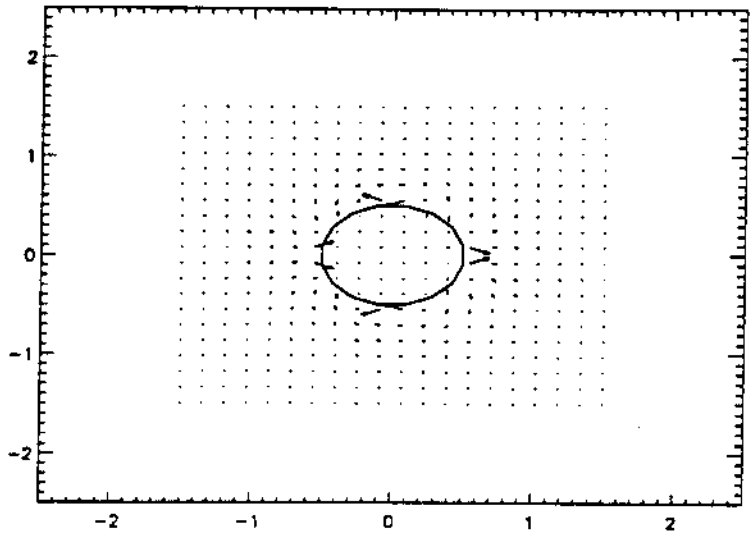


FIGURE 2. Velocity error between exact potential flow and the 16-sided approximation. The error vectors are scaled by a factor of 100 over those in Figure 1.

Finally, the method should not be considered only for viscous flows. If $\nu = 0$, there is one fewer boundary condition per segment, but one does not inject vorticity either. In other words, one solves equation (13) only with $\gamma_i = 0$. This amounts to solving the Poisson equation around a general boundary by arranging integrably weak mass sources on the boundary.

Example 1. ***Irrotational flow around a 16-sided regular polygon:*** *Perhaps one of the most popular flows to study is the flow around a cylinder. The cylinder has a diameter of one and the flow at infinity moves from left to right at a rate of one. To approximate the solution to this problem, one solves equation (13) around a 16-sided regular polygon in a uniform flow from right to left to determine the right hand side. The flow pattern induced by the line sources is plotted in Figure 1 alongside the exact solution. The error in the velocity field is shown in Figure 2.*

This method for satisfying the fluid dynamic boundary conditions is both sufficiently general and practical for large scale computations. First, there are no geometric constraints on this method. Any reasonable boundary geometry can be decomposed into a series of small plates over which flow characteristics near the boundary do not change very much. Naturally, care must be taken to use small enough segments when resolving corners and sharp edges where singularities in the potential field arise. For example, the wall jet nozzle has a 270° external angle.

By introducing continuous line sources to approximate solutions to the Poisson equation for arbitrary boundaries, the severity of the source singularity is considerably less troublesome than that of an array of point sources. Computationally, this has the advantage of reducing the stiffness of velocity integrations. Finally, all segment-segment interactions can be evaluated exactly so that numerical errors in the solving the linear system can be confined to machine precision.

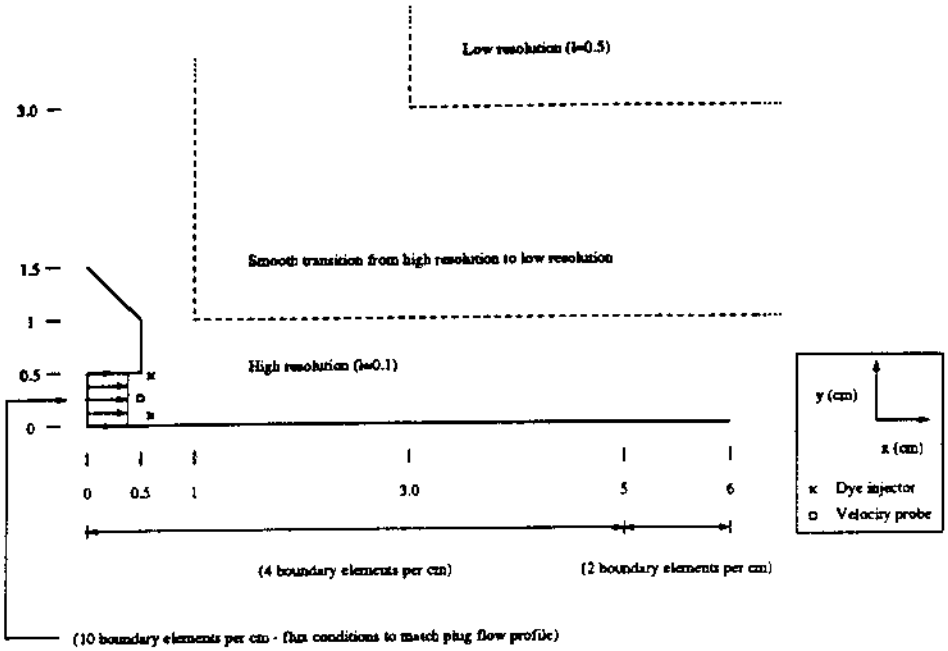


FIGURE 3. Diagram of computational domain for direct numerical simulation of wall jet flow into quiescent fluid.

4. THE WALL JET EXPERIMENTS

The numerical investigation of wall jets focuses on nozzle effects as the wall jet develops from a quiescent fluid, and the growth of boundary layer instabilities in a fully developed wall jet. The vortex method (CCSVM) together with the boundary segment technique described in this paper is used to study both problems. By applying the numerical methods directly to the startup problem, one obtains qualitative and quantitative agreement with laboratory laminar wall jet experiments in regimes where experiments and numerics can be compared. Furthermore, the transient computations reveal the existence of a long-lived vortex dipole near the nozzle. Computations of wall jet boundary layer instabilities are more subtle than the direct simulation but can remove transient and nozzle shedding effects. In this case, the boundary layer similarity solution is incorporated into the vortex method at the cost of adding a source term to equation (1).

4.1. Direct Simulation of Wall Jet Development. The experimental domain for the direct simulation of wall jet development in a quiescent fluid is specifically designed to mimic the water tank facilities of Jacob Cohen and Miki Amitay at the Technion in Haifa, Israel. Numerical parameters are chosen so that the computations will resolve the finer features of the flow near the solid wall.

The boundary segments are chosen to imitate the geometry of the water tank as illustrated in figure Figure 3. All lengths are measured in centimeters. The nozzle height is 0.5. A velocity probe, located at (0.5, 0.25) just inside the nozzle outlet, provides a measurement of vortex shedding frequency. Two clusters of dye injectors located at (0.55,0.098), (0.55,0.100), (0.55,0.102) and (0.55,0.498), (0.55,0.500), (0.55,0.502) create streaklines for flow visualization. The dye injectors

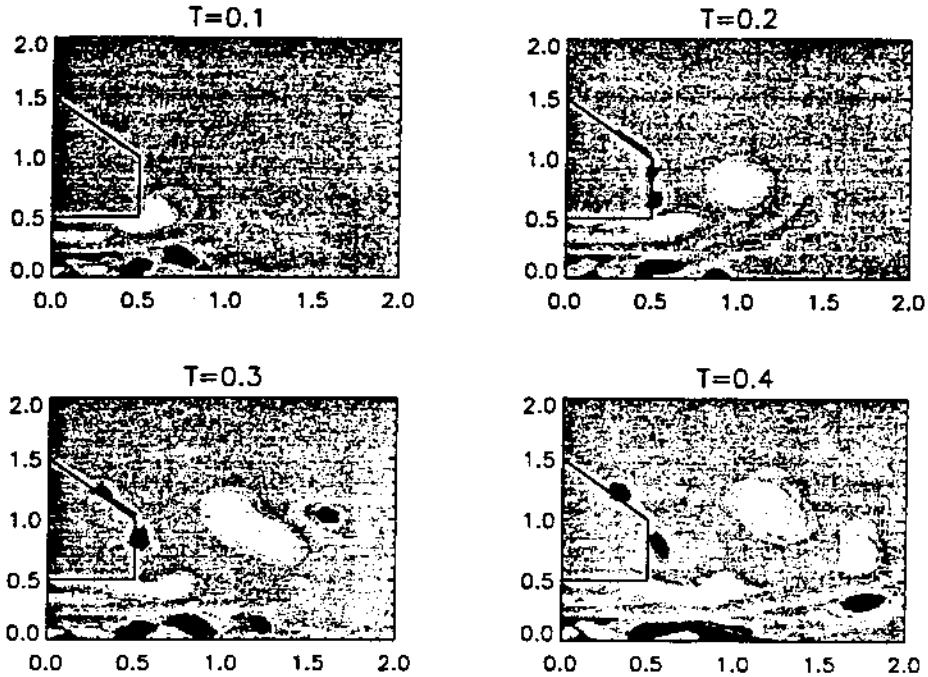


FIGURE 4. Vorticity field of R=250 wall jet during the initial formation of the boundary layer. Vorticity contours are colored in increments of 10.

introduce passive markers which convect freely with the flow velocity. The jet is created by specifying a flux at the back of the nozzle duct. All other boundaries are constrained to have zero flux and circulation. The flux distribution approximates a slug flow to match that of the water tank experiment.

As noted earlier, the boundary layer is thin compared to the extent of the dipole transient, requiring variation of the fission parameter l throughout of the domain. The parameter is small (0.1 cm) near boundaries where high resolution is required to capture dipole formation and gradually increases by a factor of five as elements move away from the boundary. High resolution simulations indicate that there is no significant error introduced by relaxing the resolution far from the wall. The fission parameter, α, is fixed at 0.7 over the entire domain. Initially, the simulation uses no vortices. As the vortices are shed from the boundary, the problem size grows, mediated only by fusion when the appropriate conditions exist. Sometime after two seconds of simulation time, the problem size exceeds 12,000 elements.

The direct wall jet simulation reveals that a large transient vortex dipole with a strong positive bias is initially shed from the nozzle into the flow. In the duration of the simulation, this dipole rotates more than 135° until the positive portion of the dipole lies directly below the negative portion as the boundary layer forms along the wall. While there has been no attempt to measure this seemingly long-lived dipole in laboratory wall jet experiments, Schneider and Zauner have observed the existence of similar vortices near plane and axisymmetric free jet nozzles, undermining the assumption that one can ignore nozzle effects and study the jet as a thin boundary layer

surrounded by irrotational fluid [14, 18]. The numerical data from these wall jet simulations suggests that the existence of this vortex may be a more generic phenomena, not restricted to free jets (see Figure 4 and Figure 7).

The dipole instability within the boundary layer immediately appears as the boundary layer is forming. Ultimately, the boundary layer separates from the solid wall at some position downstream of the nozzle and is drawn into the transient vortex. The separation region moves steadily downstream as the boundary layer establishes itself. Figure 4 portrays this initial development of the boundary layer beneath the dipole.

Most importantly, the flow visualization in the boundary layer region matches water tank experiments. Since it is not possible to directly simulate wall jet behavior indefinitely due to computational limitations, the simulated boundary layer never develops completely. However, the existing portion extends several nozzle heights downstream and can be used for comparison with experimental flow visualization. Unlike the physical wall jet experiments, the numerical data provides the full vorticity field underlying the complex dipole structures in the wall jet (see Figure 7).

4.2. Simulation of Perturbations to the Similarity Solution. Using the unstable boundary layer similarity solution of Glauert as an approximate solution to equation (1), it is possible to compute vorticity perturbations to the similarity solution. This is approach is practical and effective for two fundamental reasons. First, it is possible that the perturbations are much smaller than base flow. If this is the case, a direct simulation will have errors on the order of the vorticity of the entire flow while a simulation of the perturbations will introduce errors only on the order of the perturbation itself. Second, the base flow may automatically satisfy inhomogeneous boundary conditions over unbounded domains. If the perturbation is localized, the computational domain is reduced solely to the region of interest.

Glauert determined the following similarity solution to a wall jet boundary layer:

$$\Psi(\eta) = \nu \left(\frac{U_0}{\nu}\right)^{\frac{1}{4}} x^{\frac{1}{4}} f(\eta), \tag{33}$$

$$\eta = \frac{1}{4}\left(\frac{U_0}{\nu}\right)^{\frac{1}{4}} x^{-\frac{3}{4}} y, \tag{34}$$

where U_0 is some reference velocity, and f satisfies the ordinary differential equation

$$\begin{aligned} f''' + ff'' + 2f'^2 &= 0, \\ f(0) &= f_0, \\ f'(0) = f'(\infty) &= 0. \end{aligned} \tag{35}$$

The solution to equation (35) is solved by iterative shooting as in [5] and stored in lookup tables. Since Ψ solves boundary layer equations, it serves as an approximate solution to equation (1). If the vorticity field is decomposed as a sum of this known solution plus a perturbation, $\omega_0 + \omega'$, then equation (1) can be reduced to an evolution equation for ω':

$$\frac{D\omega'}{Dt} - \nu\nabla^2\omega' = -\vec{u}' \cdot \nabla\omega_0 \tag{36}$$

where $\frac{D}{Dt}$ retains the full velocity $\vec{u}$ and $\omega_0 = -\nabla^2\Psi$. The source term on the right can be approximated by regularly adding vortex elements to the flow on a uniform grid. The new elements

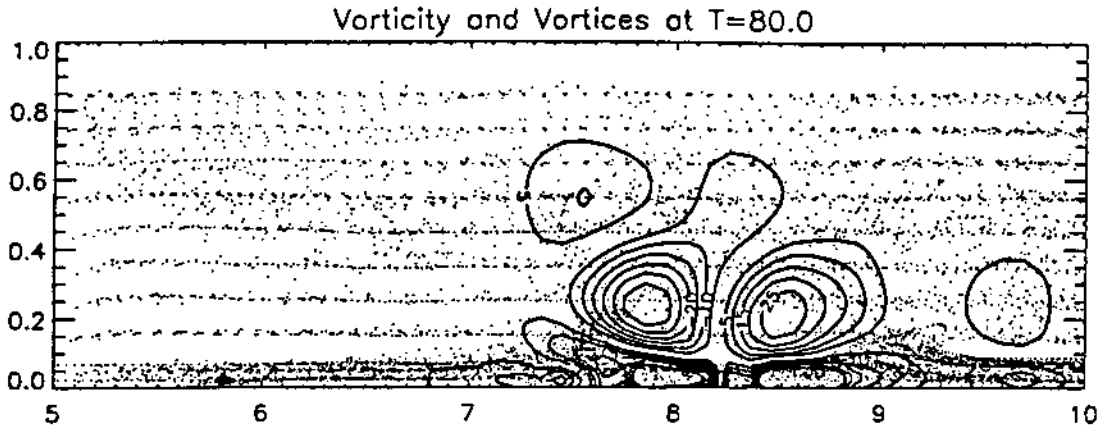


FIGURE 5. Perturbation vorticity, ω', field and computational element ("vortices") positions after 0.8 seconds for the R=250 wall jet boundary layer. The full vorticity field, $\omega_0 + \omega'$, is shown in Figure 6.

then convect with the flow as seen in Figure 5. Since ω_0 is known, the region over which elements are added can be limited to locations where $\nabla\omega_0$ is nontrivial. While this algorithm is not as adaptive as the vortex method itself, this consideration limits the region where vortices are added to a narrow band near the boundary.

The value of U_0 in equation (36) is determined by the jet flux and nozzle configuration. This is similar to free jets where the free parameter in the similarity solution is chosen based on momentum conservation. Momentum is not conserved for a wall jet, but Glauert determined another integral invariant called "external moment flux" which can be used to determine U_0:

$$F = \int_0^\infty xu \left(\int_y^\infty xu^2 dy \right) dy. \tag{37}$$

For the Re=250 wall jet described in the direct simulations, $F = \frac{125}{8}$. The external momentum flux of the Glauert solution is computed numerically for comparison. To match wall jet used in the numerical simulations, the value of U_0 is determined to be 6.25×10^6.

The computational domain for the perturbed wall jet experiments is much simpler than for the direct simulations. The domain extends from $x = 5$ to $x = 11$ spanning a distance from 10 nozzle heights to 22 nozzle heights from the boundary layer origin, far beyond the capacity of the direct simulations. The solid horizontal wall is the only boundary and is decomposed into segments of size 0.2 cm, many times the size of the dipole instabilities. Even though the computational domain of the perturbation is roughly twice as long as the direct computation, the perturbation simulation requires roughly the same number of vortices because the domain is much thinner. Where most direct simulations have broadband numerical noise to excite the most unstable mode, some noise must be added to the perturbed vortex method to excite the base flow because no vorticity is generated by the initial conditions, $\vec{u}' = 0$. To excite the flow, a small flux condition of the form

$$\delta_{\text{flux}} = \epsilon[1 - \cos(2\pi f t)] \tag{38}$$

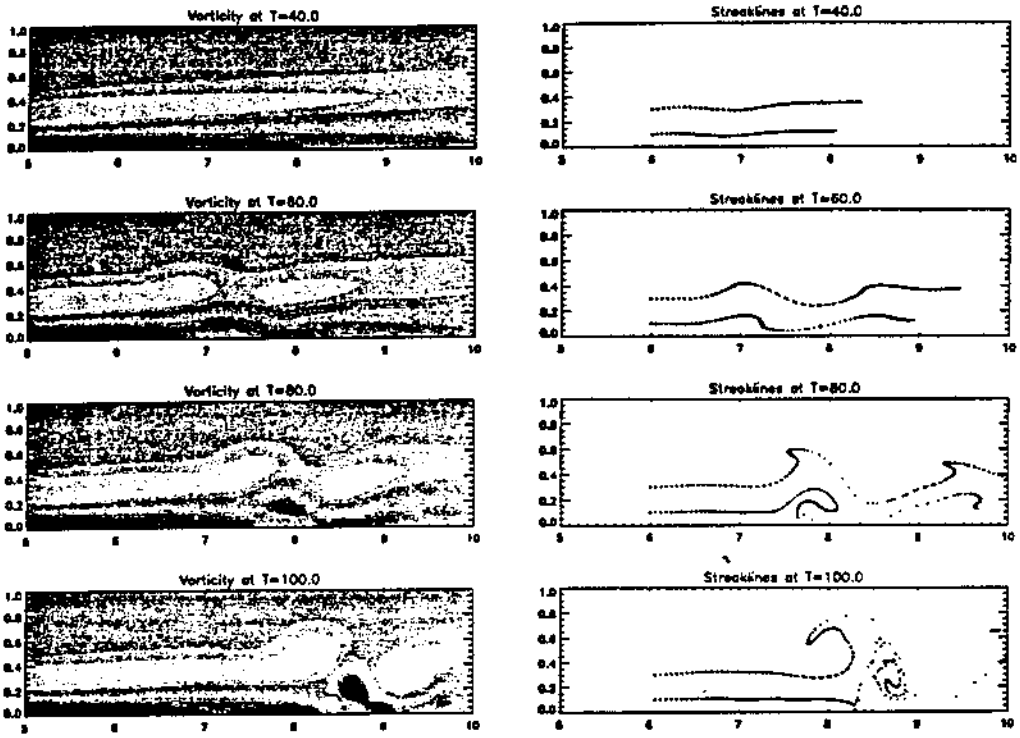


FIGURE 6. Growth of small perturbation is R=250 wall jet boundary layer. Vorticity contours are colored in increments of 10.

is imposed on the far upstream boundary segment. Values of ϵ correspond to disturbances to the velocity field of less than 1%.

Many realization of the downstream evolution of the disturbances were simulated using $f = 1$, 2 and 4. Regardless of f, all simulations exhibited the expected dipole instability at the same wavelength though there is some variation in phase. Most importantly, the simulation captures growth of minute disturbances into dipole structures (see Figure 6).

5. CONCLUSION

These simulations of wall jet flows with moderate workstation resources rely heavily upon several codependent techniques in vortex method computations. First, a new core spreading algorithm captures the viscous term of the vorticity equations with a deterministic scheme so that ensemble averages are not necessary. Since CCSVM does not rely upon operator splitting, it is possible to design methods for constraining the flow to satisfy the full boundary conditions simultaneously. Next, vortex fusion adaptively regulates problem size growth from boundary shedding and vortex fission. Some of these techniques are beyond the scope of this paper, but the concept is understandable without the technical details. A combination of these techniques make direct vortex computations of wall jet flows possible with relatively small problem sizes. Finally, by adding a source term to the vortex method, it is possible to simulate perturbations to the Glauert similarity

solution, effectively excluding nozzle effects. Not only do the simulations reveal the full nonlinear growth of disturbances, but they are more computationally efficient than direct simulations.

While these techniques are effective, there are some obvious areas for future work. First, it may be possible to boost CCSVM to higher spatial orders of accuracy using spreading, elliptical Gaussian elements. Also, injecting elliptical Gaussian elements at boundary segments is far more efficient than injecting entire arrays of circular Gaussian elements. Teng has already applied a rigid, uniform elliptical vortex method to capture the Blasius profile [17]. Analysis of this category of higher order methods requires a nonstandard convergence formalism currently under investigation by the author. Also, the Gaussian elements described here permit exact evaluation of the Biot-Savart law while elliptical Gaussians do not. In all, it is not clear that boosting CCSVM to higher spatial orders will yield faster calculations. Second, most vortex methods would benefit from fast multipole summation techniques, and CCSVM is no exception. Finally, the unsteady Rayleigh problem is merely a first order approximation for vorticity injection distributions near boundary segments. It may be possible to incorporate higher order corrections to achieve greater accuracy near boundary segments.

6. Acknowledgments

The author wishes to acknowledge the friendship and support of Bruce Bayly, Yakov Cohen and Miki Amitay. All experimental results are courtesy of Yakov Cohen (Technion at Haifa, Israel) and Miki Amitay (University of Arizona). Some of the numerical computations were performed on the Cray Y-MP/864 at the San Diego Supercomputing Center. Finally, this research is supported by NSF Grant DMS-9407660 and previously AF/OSR ASSERT Grant F496092J0324DEF.

References

1. C. R. Anderson. Vorticity boundary conditions and boundary vorticity generation for two-dimensional viscous incompressible flow. *J. Comp. Phys.*, 80:72–97, 1989.
2. G. K. Batchelor. *An Introduction to Fluid Dynamics*, pages 302–308. Cambridge University Press, 1967.
3. A. J. Chorin. Numerical study of slightly viscous flow. *J. Fluid Mech.*, 57:785–796, 1973.
4. A. J. Chorin. Vortex sheet approximation of boundary layers. *J. Comp. Phys.*, 27:428–442, 1978.
5. J. Cohen, M. Amitay, and B. J. Bayly. Laminar-turbulent transition of wall-jet flows subjected to blowing and suction. *Physics of Fluids A*, 4(2):283–289, February 1992.
6. D. Fishelov. A new vortex scheme for viscous flows. *J. Comp. Phys.*, 86(1):211–224, 1990.
7. M. B. Glauert. The wall jet. *J. Fluid Mech.*, 1:625–643, 1956.
8. C. Greengard. The core spreading vortex method approximates the wrong equation. *J. Comp. Phys.*, 61:345–348, 1985.
9. A. Leonard. Vortex methods for flow simulation. *J. Comp. Phys.*, 37:289–335, 1980.
10. L. Quartapelle. *J. Comp. Phys.*, 40:453, 1981.
11. L. Quartapelle and F. Valz-Gris. *Int. J. Num. Math. Fluids*, 1:129, 1981.
12. L. F. Rossi. Resurrecting core spreading methods: A new scheme that is both deterministic and convergent. *SIAM J. Sci. Comp.* To appear.
13. L. F. Rossi. *A Spreading Blob Vortex Method for Viscous Bounded Flows*. PhD thesis, University of Arizona, December 1993.
14. W. Schneider. Decay of momentum flux in submerged jets. *J. Fluid Mech.*, 154:91–110, 1985.
15. J. A. Sethian. A brief overview of vortex methods. In K. Gustafson and J. A. Sethian, editors, *Vortex Methods and Vortex Motion*. SIAM Frontiers in Applied Mathematics, 1990.

16. J. A. Sethian, J.-P. Brunet, A. Greenberg, and J. P. Mesirov. Two-dimensional, viscous, incompressible flow in complex geometries on a massively parallel processor. *J. Comp. Phys*, 101:185–206, 1992.
17. Z.-H. Teng. Elliptic-vortex method for incompressible flow at high Reynolds number. *J. Comp. Phys.*, 46:54–68, 1982.
18. E. Zauner. Visualization of the viscous flow induced by a round jet. *J. Fluid Mech.*, 154:111–119, 1985.

Department of Applied Mathematics, Northwestern University, 2145 Sheridan Road, Evanston, IL 60208-3111
E-mail address: rossi@jeeves.esam.nwu.edu

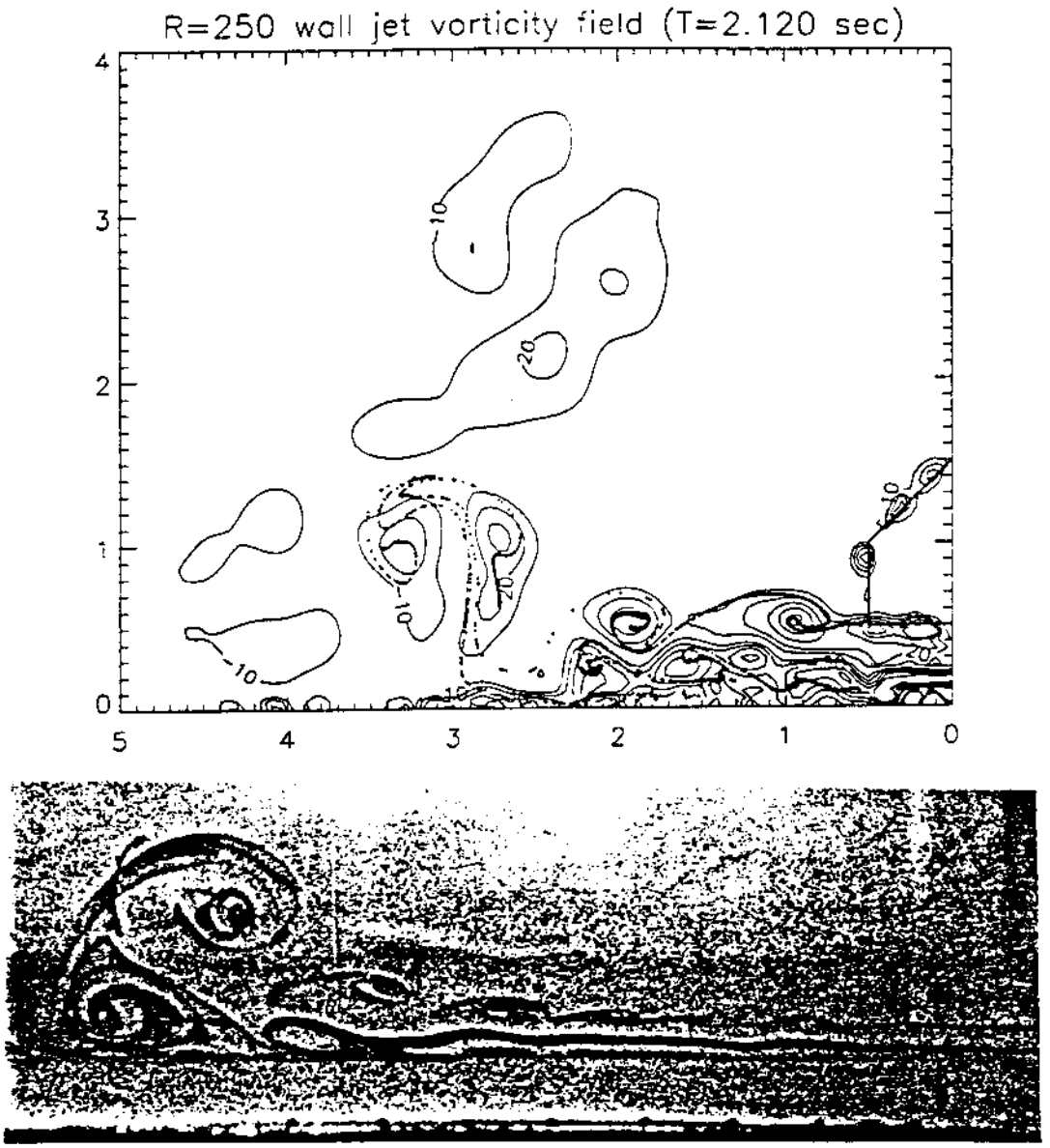


FIGURE 7. Long time development (T = 2.12 seconds) of a R=250 wall jet calculated numerically (above) is compared with R=257 water tank visualization (below). Numerical streaklines as well as vorticity contours are shown. The photographic scales of the numerical and wall jet images do not correspond, but the experimental dimensions (i.e. nozzle height) are identical. Water tank visualizations were created by Yakov Cohen and Miki Amitay at Technion in Haifa, Israel.